

Project Overview: Classification Methods in Statistical Learning

Objective:

This project explores core classification methods, their theoretical foundations, and real-world performance using the Auto dataset. The primary goals include:

- Demonstrating the quadratic decision boundary in Quadratic Discriminant Analysis (QDA)
- Illustrating the Curse of Dimensionality and its effect on local methods like K-Nearest Neighbors (KNN)
- Evaluating the accuracy of different classifiers including Logistic Regression, LDA, QDA, Naive Bayes, and KNN

1. Quadratic Discriminant Analysis (QDA): Theoretical Insight

Goal:

To prove that QDA produces quadratic decision boundaries when class variances differ.

Summary:

- In QDA, each class is modeled with its own mean and variance under a normal distribution.
- The classifier uses the posterior probability to assign class labels.
- Taking the logarithm of the probability function simplifies the discriminant function and reveals a term involving the square of the input variable.
- As a result, the comparison between class probabilities becomes a quadratic function, confirming that QDA decision boundaries are curved, not linear.
- **Conclusion:** QDA forms quadratic boundaries unless class variances are identical, in which case it reduces to Linear Discriminant Analysis (LDA).

2. Curse of Dimensionality: KNN Limitations

Goal:

To illustrate how increasing feature dimensionality drastically reduces the effectiveness of local methods.

Dimensional Cases:

- 1 Dimension ($p = 1$): About 10% of the data is near a test point.
- 2 Dimensions ($p = 2$): Only 1% of the data is nearby.
- 100 Dimensions ($p = 100$): Almost no data points are close enough to be useful.

Implications:

- As the number of features grows, the data becomes sparse.
- In high dimensions, the “neighborhood” around a point expands drastically, requiring nearly the entire feature space to capture even a small percentage of data.
- **Conclusion:** Local methods like KNN become unreliable in high-dimensional settings, as they rely on the notion of "closeness," which breaks down in large feature spaces.

3. Logistic Regression: Estimating Probabilities**Use Case:**

Estimate the probability that a student receives an A in a statistics course based on:

- Hours studied
- Undergraduate GPA

Findings:

- A student studying 40 hours with a 3.5 GPA has about a 38% chance of receiving an A.
- To achieve a 50% chance, the student would need to study 50 hours.
- **Conclusion:** Logistic regression provides interpretable probabilistic insights into how input features influence binary outcomes.

4. Auto Dataset Analysis with Classification Models**Dataset Summary:**

- Used the Auto dataset after removing missing values and irrelevant columns.
- Created a binary variable mpg01 (1 if miles per gallon is above the median, 0 otherwise).
- Key predictors used: displacement, horsepower, weight, and year.

Model Evaluation Results (70% Train, 30% Test Split):

Model	Test Error (%)	Notes
Linear Discriminant Analysis (LDA)	9.32	Linear boundaries, good baseline performance
Quadratic Discriminant Analysis (QDA)	11.86	Slightly worse than LDA, more flexible
Logistic Regression	11.02	Best among parametric models
Naive Bayes	11.02	Comparable to Logistic Regression
K-Nearest Neighbors (KNN)	3.39	Best overall with optimal $k = 4$

Visual Insights:

- High mpg cars tend to have lower displacement, horsepower, and weight, and are more recent models.
- Boxplots and scatterplots confirmed these associations.

Key Takeaways:

- QDA offers flexible modeling but requires unequal variances for full advantage.
- The curse of dimensionality limits the usefulness of local algorithms like KNN in high-dimensional data.
- Logistic regression effectively estimates binary outcomes and allows for threshold analysis.
- KNN was the best-performing model on the Auto dataset but may not scale well to more complex, high-dimensional tasks.

Conclusion:

This project blends theoretical understanding with empirical evaluation, demonstrating how model assumptions affect both performance and interpretability. While KNN excelled in this case, the findings emphasize the importance of data dimensionality, variable selection, and model fit when choosing classification techniques for real-world problems.